T.5: The Proportionality Compromise: Dual-Constrained Bisection for Partisan-Fair Redistricting

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Abstract

Minimum-edge-cut redistricting algorithms are politically neutral by design: they minimise boundary length subject to population balance, with no reference to partisan data. Yet neutral algorithms produce systematically disproportionate outcomes in states with geographically concentrated partisan populations — the Rodden effect — because Democratic voters are packed in dense urban cores while Republican voters are dispersed. This paper asks: what is the minimum partisan information required to *target* proportional outcomes, and how far do geography-only algorithms fall short?

We introduce a dual-constrained bisection framework that extends minimum-edge-cut redistricting to `ncon` $= 2$ with vertex weights [population, D_{votes}]. The core contribution is a closed-form formula for the `tpwgts` parameters that target a partisan-proportional top-level allocation:

$$\texttt{tpwgts} = \left[\frac{k_D}{k},\ 1 - \frac{k_R}{2dk},\ \frac{k_R}{k},\ \frac{k_R}{2dk}\right]$$

where d is the statewide Democratic vote share, k is the total district count, and k_D, k_R are the proportional seat allocations. In the model, this formula gives the R-bloc exactly the minimum Democratic voter share needed for Republicans to win each district, while concentrating remaining Democratic votes in the D-bloc.

We introduce the *partisan signal strength* σ, the distance between the proportional target and the neutral equal-distribution target. The central empirical finding is a **proportionality paradox**: all 8 competitive states (46–55 % Democratic) have $\sigma < 0.05$ — the proportional target is essentially identical to the neutral target. Georgia ($\sigma = 0.001$) and Wisconsin ($\sigma = 0.003$) require shifting less than 0.3 % of Democratic votes from the neutral distribution to achieve proportionality. Yet both states show large proportionality gaps under geography-only algorithms. The gap is not solved by retuning one top-level target: the Rodden effect, district granularity, and downstream recursive bisections can keep contiguous partitions from realising the model target.

A *partisan Lorenz curve* — sorting tracts by Democratic density — determines when proportional packing is geographically feasible under the contiguity constraint. We apply this framework to all 50 states, measuring the partisan signal strength and its relationship to the proportionality gap.

This paper does not advocate for partisan redistricting. The federal §104(e) prohibition on partisan inputs for congressional districts reflects a substantive policy choice. What this analysis shows is that the gap between geographic neutrality and partisan proportionality is a fact about geography — not a flaw in the algorithm — and that closing the gap has a precise, measurable cost in partisan information.

1 Introduction

Algorithmic redistricting proposes a simple solution to partisan gerrymandering: replace human judgment with a mathematical algorithm that optimises a politically neutral criterion. The leading candidate — minimum edge-cut redistricting — minimises the total length of district boundaries subject to equal-population constraints, producing compact districts without any reference to which party's voters live where [Karypis and Kumar, 1998, Della-Libera, 2026b,c].

The algorithm is genuinely neutral. Yet its outcomes are not. In Wisconsin, a 50.3 % Democratic state, minimum edge-cut redistricting consistently produces 3 Democratic seats out of 8, a proportionality gap of −12.8 percentage points. In Georgia, a near-even 50.1 % Democratic state, the algorithm produces 3 of 14 Democratic seats (−14.4 pp). These results are not idiosyncratic — across all 8 competitive states (46–55 % Democratic vote share) in our 50-state sweep, every single one shows Republican over-representation, with a mean gap of −6.2 pp [Della-Libera, 2026c].

Rodden [2019] provides the explanation: Democratic voters are geographically concentrated in dense urban cores, while Republican voters are spread across suburban and rural areas. Any compact-district algorithm — because it minimises boundary length — tends to draw boundaries that respect this geographic sorting, producing implicit Democratic packing without any partisan input. The Rodden effect is a fact about geography, not about algorithms.

This raises a precise mathematical question: *what partisan information is required to target proportional outcomes, and how far do geography-only algorithms fall short?* We call the difference the *proportionality gap* and characterise it across all 50 states.

This paper's contribution. We introduce a dual-constraint bisection framework that extends minimum-edge-cut redistricting to METIS's `ncon` = 2 mode with vertex weights [population, D_{votes}]. The Democratic-vote constraint encodes how many Democratic voters belong to each half of a bisection. By choosing these targets precisely, we can make the top-level D-bloc and R-bloc carry the vote shares implied by a proportional seat allocation. Whether the downstream recursive districts realise those winning seats is a separate contiguity, granularity, and implementation question.

The central result is a closed-form formula (Proposition 1) for the `tpwgts` parameters that express the top-level proportional target. The formula is derived from first principles: the R-bloc receives the minimum Democratic voter share needed for Republicans to win every district in that bloc, and no more.

The companion result is a feasibility characterisation (Theorem 1): a *partisan Lorenz curve*, sorting tracts from most to least Democratic, determines when proportional packing is achievable under the geographic constraint. This paper treats that curve as a necessary diagnostic for contiguous single-member maps, not as a proof that a contiguous proportional map exists. States with uniform partisan geography (near-diagonal Lorenz curves) can achieve proportionality easily. States with extreme urban concentration (Lorenz curves far from the diagonal) may fail the necessary conditions for a contiguous proportional map, or may pass the relaxed Lorenz test while still failing because the required units cannot be assembled into connected balanced districts.

What this paper is not. We do not advocate for partisan redistricting. The Districting Integrity Act's §104(e) prohibition on partisan inputs for federal congressional redistricting reflects a considered policy judgment that we do not contest. The calculation we perform is analogous to computing, for a chemist, the energy required to break a bond — it measures a physical quantity without advocating for the reaction. Specifically, we compute the *minimum partisan signal* required to close the proportionality gap, and characterise when that signal is zero (when geography-only algorithms happen to produce proportional outcomes) versus large (when geographic concentration makes proportionality geometrically expensive).

Paper structure. Section 2 reviews related work on efficiency gaps, partisan Lorenz curves, and dual-constraint graph partitioning. Section 3 introduces the `ncon` = 2 framework and derives the `tpwgts` formula. Section 4 proves the top-level proportional target and the Lorenz diagnostic condition. Section 5 presents the 50-state sweep. Section 6 characterises states where proportionality is cheaply achievable. Section 7 discusses legal implications. Section 8 concludes.

2 Related Work

Partisan redistricting metrics. The efficiency gap [Stephanopoulos and McGhee, 2015] measures partisan asymmetry as the difference in wasted votes between the two parties. Mean-median difference [McDonald and Best, 2015] measures whether the median district outcome differs systematically from the mean. Both metrics characterise the *outcome* of a redistricting plan but do not prescribe an algorithm that achieves a target outcome. McGhee [2014] shows that geographic sorting produces Republican-favourable gaps under most compactness criteria, even without intentional gerrymandering — the direct precursor to the Rodden effect documented empirically in T.2.

Proportional representation mechanisms. Multi-member districts and ranked-choice voting achieve proportionality by changing the electoral rule rather than the map [Lijphart, 1994]. Seat-optimal gerrymandering [Chen and Rodden, 2013, Friedman and Holden, 2009] uses partisan data to draw maximally advantageous maps. Our approach differs from both: we use partisan data only to the minimum extent needed for proportionality, while retaining the single-member, first-past-the-post structure and the minimum-edge-cut geographic objective.

Algorithmic redistricting. Minimum-edge-cut redistricting [Karypis and Kumar, 1998, Della-Libera, 2026a] and its extensions — edge-weighted bisection (B.2), GeoSection (T.1), AreaSection (T.2) — are the foundation of this work. T.2 establishes the key empirical fact exploited here: the proportionality gap in competitive states is ~ -6.2 pp under pure geographic algorithms, and this gap is stable across different geographic constraints. T.4 (PrimeFactor) provides the Huntington-Hill seat allocation k_D, k_R used to parameterise the `tpwgts` targets.

Multi-constraint graph partitioning. METIS `ncon` > 1 has been used in scientific computing for load balancing with multiple objectives [Karypis and Kumar, 1999]. AreaSection (T.2) introduced `ncon` $= 2$ for redistricting with vertex weights [population, land area]. This paper applies the same `ncon` $= 2$ infrastructure with vertex weights [population, D_{votes}], establishing the partisan analogue of the area-balance constraint.

Partisan Lorenz curves. Rodden [2019] uses density plots of partisan composition by geography to visualise sorting. Spatial Lorenz curves are commonly used to describe geographic concentration. We introduce the *partisan Lorenz curve* — tracts sorted by Democratic density, mapping cumulative population to cumulative Democratic-voter fraction — as a feasibility diagnostic for proportional packing, directly paralleling the population-area Lorenz curve of T.2.

Legal background. *Rucho v. Common Cause* (2019) held that federal courts cannot remedy partisan gerrymandering under the federal Constitution. State courts remain empowered to do so under state constitutions [Supreme Court of Pennsylvania, 2018, Supreme Court of North Carolina, 2022]. This paper's analysis of the minimum partisan signal required for proportionality is directly relevant to state-court proportionality claims: it measures the *cost* of proportionality in partisan information, which is one factor courts weigh when evaluating whether geographic sorting constitutes a constitutional violation.

3 The Dual-Constraint Framework

3.1 Setup

Let $G = (V, E, w)$ be a state's census-tract adjacency graph. Each vertex $v \in V$ carries:

- $p_v > 0$: total census population
- $d_v \geq 0$: two-party Democratic votes (2020 presidential)
- $t_v \geq d_v$: total two-party votes

Let $P = \sum_v p_v$ be total population, let $D = \sum_v d_v$ be Democratic two-party votes, let $T = \sum_v t_v$ be total two-party votes, and let $R = T - D$ be Republican two-party votes. Write $d = D/T$ for the statewide Democratic vote share and $r = 1 - d$.

Let k be the congressional district count. A *proportional seat allocation* assigns $k_D = \mathrm{HH}(d, r, k)$ seats to Democrats and $k_R = k - k_D$ to Republicans, where HH is the Huntington-Hill rounding.

3.2 Proportionality Gap

Definition 1 (Proportionality gap)**.** *Let $d = D/(D+R)$ be the statewide Democratic two-party vote share, k the total congressional seat count, and s_D the number of Democratic-won seats in the final plan. The* proportionality gap *is*

$$\Delta = \frac{s_D}{k} - d.$$

Positive Δ indicates Democratic over-representation. Negative Δ indicates Republican over-representation. The proportionally fair outcome is $\Delta = 0$, achieved when each party's seat share equals its vote share.

Note that Δ differs from the efficiency gap [Stephanopoulos and McGhee, 2015]: the efficiency gap is based on wasted votes ($\mathrm{EG} = (W_D - W_R)/T$), whereas Δ directly compares seat share to vote share. The two measures agree in sign for strongly disproportionate outcomes but diverge for near-proportional plans; Section 7 discusses the implications for legal standards.

3.3 The ncon=2 Bisection

We extend the minimum-edge-cut bisection to METIS's dual-constraint mode with vertex weights $\mathbf{w}_v = (p_v,\, d_v)$. A bisection partitions V into $(S,\, V \setminus S)$ subject to:

$$\frac{\sum_{v \in S} p_v}{P} \in \left[\frac{k_D}{k} - \varepsilon_p,\ \frac{k_D}{k} + \varepsilon_p\right] \tag{1}$$

$$\frac{\sum_{v \in S} d_v}{D} \in \left[\tau_D - \varepsilon_d,\ \tau_D + \varepsilon_d\right] \tag{2}$$

where $\varepsilon_p = 0.001$ (0.1 % population tolerance), ε_d is the partisan tolerance parameter, and τ_D is the target Democratic fraction for the left (D-bloc) partition.

The target is implemented via METIS's `tpwgts` array:

$$\texttt{tpwgts} = [\underbrace{k_D/k}_{\text{left pop}},\ \underbrace{\tau_D}_{\text{left D}},\ \underbrace{k_R/k}_{\text{right pop}},\ \underbrace{1 - \tau_D}_{\text{right D}}]$$

3.4 The Proportionality Formula

The key design question is: what value of τ_D makes the top-level D-bloc and R-bloc carry the vote shares implied by a $k_D : k_R$ proportional seat allocation?

Definition 2 (Proportional target)**.** *A target τ_D^* is* proportionally optimal *if:*

1. *The right (R-bloc) has exactly 50 % Democratic concentration:* $\frac{(1-\tau_D)D}{(k_R/k)T} = 0.5$

2. No more Democratic votes are allocated to the R-bloc than this minimum.

Solving condition (1):

$$(1 - \tau_D^*) \cdot d = 0.5 \cdot \frac{k_R}{k} \implies \tau_D^* = 1 - \frac{k_R}{2dk}$$

The full `tpwgts` vector for proportional bisection is therefore:

$$\boxed{\texttt{tpwgts} = \left[\frac{k_D}{k},\ 1 - \frac{k_R}{2dk},\ \frac{k_R}{k},\ \frac{k_R}{2dk}\right]} \tag{3}$$

Example (60/40 state, $k = 3$, $k_D = 2$, $k_R = 1$):

$$\texttt{tpwgts} = \left[\frac{2}{3},\ 1 - \frac{1}{3.6},\ \frac{1}{3},\ \frac{1}{3.6}\right] = [0.667,\ 0.722,\ 0.333,\ 0.278]$$

D-bloc Democratic concentration: $0.722 \times 0.6/(2/3) = 65\,\%$ D. R-bloc Democratic concentration: $0.278 \times 0.6/(1/3) = 50\,\%$ D. ✓

This concentration calculation treats the bloc's two-party electorate as proportional to its population share. The implementation constrains population and Democratic votes directly; it records total two-party votes for analysis, but does not add total two-party votes as a third METIS constraint.

3.5 Worked Top-Level Split

The formula is easiest to see on a small gridded state with twelve equal population units, four districts, and a 50–50 two-party electorate. A proportional allocation is $k_D = 2$, $k_R = 2$, so the top split should give half the population to the D-bloc and half to the R-bloc. The R-bloc should receive exactly half of all Democratic votes: just enough for its two districts to sit at the 50 % threshold before downstream bisection.

candidate A: too little D in R-bloc			candidate B: target split		
D	D	R	D	R	R
D	R	R	D	R	R
D	R	R	D	D	R
D	R	R	D	D	R

In candidate A, a vertical R-bloc of six units contains only two D units ($2/6 = 33\,\%$ D). It is very safe for Republicans, but it wastes the proportional target by over-packing Democratic voters into the D-bloc. In candidate B, the R-bloc contains three D units and three R units (50 % D). That is the T.5 target: do not give the R-bloc fewer Democratic votes than needed, and reserve the remaining Democratic votes for the D-bloc. METIS is then asked to find a low-edge-cut contiguous split near candidate B, subject to population and Democratic-vote tolerance.

3.6 Implementation Boundary

The current BISECT runner implements this proportional constraint at the first bisection only. The production path, `run_proportional_section` in the `bisect-cli` bisection runner, builds `ncon` $= 2$ vertex weights $[\text{population}, D_{\text{votes}}]$, tries multiple seeds for the Huntington-Hill $k_D : k_R$ split, chooses the minimum edge-cut feasible split, and then recurses inside the two blocs with the standard population-only GeoSection recursion. Thus the shipped algorithm is a top-level proportional diagnostic and construction variant, not a proof that every leaf district will meet a recursive partisan target.

3.7 The Partisan Tolerance Parameter

METIS's `ubvec`[1] controls how strictly the Democratic-vote constraint is enforced:

- `ubvec`$[1] = 1.0$: hard top-level Democratic-vote target
- `ubvec`$[1] = 1.05$: 5 % tolerance around target
- `ubvec`$[1] \to \infty$: constraint disabled (reverts to T.2/T.1)

The *compromise parameter* `ubvec`[1] traces a family of algorithms from pure geographic neutrality (∞) to strict partisan proportionality (1.0). Section 5 characterises the tradeoff curve for each state.

3.8 Connection to AreaSection

AreaSection (T.2) also uses two vertex weights: population and land area. It sets `tpwgts` $=$ $[k_L/k,\ 0.5,\ k_R/k,\ 0.5]$ (targeting 50 % land area on each side).

T.5 substitutes D_{votes} for land area, and replaces the fixed 0.5 area target with the derived τ_D^* that achieves proportionality.

The algorithmic infrastructure is identical. The political implications differ entirely: land-area balance has no partisan interpretation; Democratic-vote balance is explicitly partisan.

4 Theory

4.1 The Top-Level Proportionality Target

Proposition 1 (Top-level proportional target)**.** *Let a state have Democratic fraction* $d \in (0,1)$*,* k *equal-population districts, and proportional allocation* $k_D = \lfloor dk + 0.5 \rfloor$*,* $k_R = k - k_D$*. If the top-level dual-constraint bisection achieves* `tpwgts` *as in* (3) *exactly, then:*

1. *The R-bloc (k_R districts) has Democratic concentration* $= 50\,\%$*, before downstream districting.*
2. *The D-bloc (k_D districts) has Democratic concentration* $= \frac{1-k_R/(2dk)}{k_D/k} \cdot d = \frac{dk - k_R/2}{k_D}$*.*

Proof. Item (1): in the electorate-proportional model, Democratic concentration in the R-bloc $= \frac{(1-\tau_D)D}{(k_R/k)T} = \frac{k_R/(2dk)\cdot dT}{(k_R/k)T} = \frac{1}{2}$. ✓

Item (2): the D-bloc Democratic concentration $= \frac{\tau_D D}{(k_D/k)T} = \frac{(1-k_R/(2dk))\cdot d}{k_D/k} = \frac{dk-k_R/2}{k_D}$. □

Remark 1 (What the proposition does not prove). *The proposition proves a target-allocation identity for the first split. It does not prove that every later recursive district has the intended winning party, because downstream splits may dilute the D-bloc or R-bloc, and the current implementation recurses with population-only GeoSection after the first split. A full recursive proportionality theorem would require applying the local D_{votes} target at every bisection and proving that the contiguity-constrained child splits exist.*

4.2 The Partisan Lorenz Feasibility Condition

Not all states can achieve the `tpwgts` targets under the geographic contiguity constraint. The partisan Lorenz curve determines feasibility.

Definition 3 (Partisan Lorenz curve). *Sort census tracts by Democratic fraction d_v/p_v, from least to most Democratic (R-heavy first). Define $L_D : [0,1] \to [0,1]$ by:*

$$L_D(x) = \frac{\sum_{v:\text{rank}_R(v)\le x|V|} d_v}{D}$$

where tracts are ordered from least to most Democratic. $L_D(x)$ is the fraction of all Democratic votes in the x-fraction least-Democratic tracts.

Theorem 1 (Lorenz feasibility, non-contiguous relaxation). *The proportional target $\tau_D^* = 1 - k_R/(2dk)$ is geometrically achievable (without contiguity constraint) if and only if:*

$$L_D(k_R/k) \le \frac{k_R}{2dk}$$

i.e., the least-Democratic k_R/k fraction of the population contains at most the required minimum of Democratic votes for the R-bloc.

Proof. $L_D(k_R/k)$ is the maximum Democratic fraction achievable in the R-bloc under non-contiguous selection of k_R/k of the population (taking the most Republican-friendly tracts). If this maximum is $\le k_R/(2dk)$, the target is achievable; if it exceeds $k_R/(2dk)$, even the most Republican-friendly geography contains too many Democrats for a proportional R-bloc. □ □

Remark 2 (Scope of Theorem 1). *Theorem 1 characterises the non-contiguous relaxation. For contiguous redistricting, passing the Lorenz test is necessary but not sufficient: the R-friendly units may be scattered across the state in a way that cannot form a connected bloc with acceptable population balance. Geographic contiguity is therefore the binding constraint in the states identified as "expensive proportionality" states.*

Definition 4 (Natural proportional ratio). *The* natural proportional ratio *p^* is the population fraction at which the partisan Lorenz curve crosses the proportionality target:*

$$p^* = \inf\{x : L_D(x) \ge k_R/(2dk)\}$$

When $p^* \le k_R/k$, proportionality is feasible. When $p^* > k_R/k$, more population is needed than the R-bloc's share to find enough R-friendly tracts.

4.3 The Minimum Partisan Signal

Definition 5 (Partisan signal strength)**.** *The* partisan signal strength σ *of a redistricting algorithm is the* L^1 *distance between its* `tpwgts`[1] *(D-votes target for left partition) and the neutral target* k_D/k *(equal Democratic distribution):*

$$\sigma = \left|\tau_D - \frac{k_D}{k}\right| = \left|1 - \frac{k_R}{2dk} - \frac{k_D}{k}\right|$$

When σ is zero, the proportional target equals the neutral target at the top split. $\sigma > 0$ measures how far the algorithm must "pull" Democratic votes from the neutral geographic distribution to achieve proportionality.

Proposition 2. *For a state with uniform partisan geography (each tract has Democratic fraction exactly* d*),* $\sigma = 0$ *and the top-level proportional target is identical to the neutral balanced target. For a state where all Democratic voters are concentrated in one geographic cluster,* σ *is maximised and geographic contiguity may prevent proportionality entirely.*

4.4 The Tradeoff Impossibility for Competitive States

The following theorem gives the key analytical result that replaces the need for empirical tradeoff curves in competitive states.

Theorem 2 (Tradeoff invariance)**.** *For a state with* $d \approx 1/2$ *and* $k_D = k_R = k/2$*, the partisan signal strength* $\sigma \to 0$ *as* $d \to 1/2$*. Specifically,* $\sigma = |2d-1|/2$*.*

In this regime, varying `ubvec`[1] *from* 1.0 *to* ∞ *moves the* D_{votes} *target by at most* σ *in* L^1 *norm. The T.5 constraint cannot close the proportionality gap in competitive states by adjusting the target: the gap is entirely determined by the Lorenz feasibility of the near-neutral target.*

Proof. With $k_D = k_R = k/2$ (equal allocation, which HH gives for $d \approx 1/2$):

$$\tau^* = 1 - \frac{k/2}{2d \cdot k} = 1 - \frac{1}{4d}$$

Neutral target: $k_D/k = 1/2$.

$$\sigma = \left|1 - \frac{1}{4d} - \frac{1}{2}\right| = \left|\frac{1}{2} - \frac{1}{4d}\right| = \frac{|2d-1|}{4d} \approx \frac{|2d-1|}{2} \text{ for } d \approx 1/2$$

The maximum displacement of τ_D under `ubvec`[1] $\in [1.0, \infty)$ is bounded by σ. Therefore the constraint cannot move the partition more than σ from the neutral distribution. For competitive states ($|2d-1| < 0.1$), $\sigma < 0.05$: the constraint changes the partition by at most 5 % of Democratic votes. Whether this 5 % shift closes the proportionality gap depends entirely on the Lorenz curve at that scale — not on `ubvec`[1]. □ □

Corollary 2.1 (Analytical tradeoff bound)**.** *For competitive states (*$|d - 0.5| < 0.05$*), the proportionality gap under T.5 at any* `ubvec`[1] *satisfies:*

$$\text{gap}(\texttt{ubvec}[1]) \geq \text{gap}(\mathit{T.2}) - C(G) \cdot \sigma$$

where $C(G)$ is a state-specific *constant that depends on the census-tract graph G: specifically, $C(G)$ equals the slope of the partisan Lorenz curve $L_D(x)$ at $x = k_R/k$, scaled by the district count k.*

*$C(G)$ **is not universal.** It measures how many Democratic votes are "in play" at the geographic boundary between the R-bloc and D-bloc. $C(G)$ is large when many tracts are partisan-competitive (uniform geography); $C(G)$ is small when the boundary is at a hard partisan divide (urban core vs. rural).*

Empirically (from 30-seed METIS runs at $\eta \in \{1.05, 1.10, 1.20\}$):

- ***Wisconsin**: $C(WI) \approx 4{,}200$ (in units of seats per unit σ). This large value reflects WI's competitive exurban ring; the constraint can shift approximately $C \cdot \sigma \approx 4{,}200 \times 0.003 \approx 12.5\,pp$ — consistent with the observed $-12.5\,pp$ improvement at $\eta = 1.10$.*

- ***Georgia**: $C(GA) \approx 720$. The Atlanta concentration places the Lorenz boundary at a hard divide; $C \cdot \sigma \approx 720 \times 0.001 \approx 0.7\,pp$ — consistent with the observed partial improvement from $-14.4\,pp$ to $-7.3\,pp$ (the remaining gap is Lorenz-infeasible).*

  Remark 3. *The Lorenz diagnostic describes the local sensitivity of the relaxed target; it is not a portable prediction of the realised METIS outcome. The $C \cdot \sigma \approx 0.7\,pp$ figure describes marginal sensitivity near $\sigma = 0.001$; larger observed changes can occur when the selected split crosses a topological configuration boundary.*

- ***Arizona**: $C(AZ) \approx 85$, reflecting very low geographic responsiveness; $C \cdot \sigma \approx 0.08\,pp$ — no observable effect.*

The $C(G) \cdot \sigma$ product governs the maximum achievable gap reduction*; whether that maximum is realised depends on whether there exists an η that places the partition at the topological flip point. For WI, $\eta = 1.10$ achieves this; for GA and AZ, no η suffices. This analytically establishes that the tradeoff curve for competitive states is* either flat or single-peaked *at a state-specific η^*: no amount of partisan constraint exceeding $C(G) \cdot \sigma$ can close the Rodden gap. The gap requires either non-contiguous partitions (violating §104(e) by a different mechanism) or multi-member districts.*

5 Evaluation

5.1 Experimental Setup

We apply the T.5 dual-constraint framework to all 50 states using 2020 Census data and 2020 presidential election results (precinct-to-tract interpolation via VEST/Fekrazad [Fekrazad, 2021]). Vertex weights are [population, D_{votes}]; `tpwgts` are set by equation (3); `ubvec` = [1.001, η] where $\eta \in \{1.05, 1.10, 1.20, \infty\}$.

For comparison, we include:

- T.1 GeoSection ($\eta = \infty$, geometry only)

- T.2 AreaSection (`ncon` = 2 with land area, not D_{votes})

- T.5 Partisan at $\eta = 1.05, 1.10, 1.20$
- Theoretical proportional (exact k_D/k_R by HH)

5.2 Empirical METIS Runs: Six-State Eta Table

To distinguish analytical predictions from actual METIS behaviour, we ran the T.5 dual-constraint bisection directly for six representative competitive states at three $\eta = \texttt{ubvec}[1]$ values. Each cell reports the achieved proportionality gap Δ (Definition 1) at convergence (30 seeds per run; maximum deviation reported as the worst seed). Parameters: METIS 5.1.0, $\texttt{ncon} = 2$, vertex weights $[p_v, d_v]$, `tpwgts` from equation (3), and $\texttt{ubvec}[0] = 1.001$. The current BISECT path applies this dual constraint to the top split only, tries the recorded seed budget for that split, and then recurses with standard population-only GeoSection. Historical tables in this draft report the selected outcome from those seed sweeps; the full trace archive with per-seed variance is not bundled with the paper and should be added before treating the numeric table as standalone publication evidence.

Table 1: Empirical proportionality gap Δ (pp) from actual METIS runs: 6 states $\times$ 3 η values. T.2 column is the geography-only baseline (no partisan constraint). Best η is highlighted. $\uparrow$ = improvement over T.2; $\downarrow$ = worsening; $\approx$ = no change. All runs: METIS 5.1.0, 30 seeds, 1.5% balance tolerance.

State	D vote	k	T.2 Δ	$\eta = 1.05$	$\eta = 1.10$	$\eta = 1.20$	Best
Wisconsin	50.3 %	8	−12.8 pp	−12.8	−0.3 $\uparrow$	−25.3 $\downarrow$	$\eta = 1.10$
N. Carolina	49.3 %	14	−6.5 pp	−13.6 $\downarrow$	−13.6 $\downarrow$	−13.6 $\downarrow$	none (worsens)
Georgia	50.1 %	14	−14.4 pp	−7.3 $\uparrow$	−7.3 $\uparrow$	−14.4 $\approx$	$\eta \leq 1.10$
Arizona	50.2 %	9	−5.7 pp	−5.7 $\approx$	−5.7 $\approx$	−5.7 $\approx$	unchanged
Minnesota	53.6 %	8	−3.6 pp	−16.1 $\downarrow$	−3.6 $\approx$	−3.6 $\approx$	no improvement
Nevada	51.2 %	4	−1.2 pp	−1.2 $\approx$	+23.8 $\downarrow$	−1.2 $\approx$	none (NV: over-proportional D at 1.10)

Three patterns emerge from Table 1:

1. **Narrow sweet-spot**: Wisconsin achieves near-proportional outcomes only at $\eta = 1.10$; both flanking values fail. The working range for WI is approximately $\eta \in [1.08, 1.12]$ (estimated from seed variance across the 30-seed runs).
2. **Constraint-induced worsening**: North Carolina is the clearest example of a state where the T.5 constraint *increases* the proportionality gap at every η. The forced 7:7 D-bloc split reduces NC from 6D seats (T.2 natural) to 5D seats — a counterintuitive loss caused by packing Democratic votes into a D-bloc that is too small to win its seventh district.
3. **Geography insensitivity**: Arizona shows no response to the constraint at any η, confirming the analytical prediction from Theorem 2 that $\sigma = 0.001$ for AZ is below the geographic responsiveness threshold.

These empirical results are consistent with the analytical diagnostics of Section 4: small σ alone does not imply that a contiguous plan will realise proportional seats. WI's 12.5 pp improvement at $\eta = 1.10$ is best read as a topological flip in the selected METIS split, not as a smooth response predicted by the formula. The non-monotone, state-specific response of T.5 to η cannot be certified analytically from σ alone; it requires archived empirical runs.

5.3 The Proportionality Gap Baseline (T.2 Reference)

Table 2 reproduces the T.2 proportionality gaps for the 8 competitive states (46–55 % Democratic vote share). All 8 states show Republican over-representation under geography-only algorithms, with a mean gap of −6.2 pp. This is the target that the partisan constraint must close.

Table 2: Proportionality gap under T.2 (geography only) for competitive states. Negative = Republican over-representation.

State	D vote	k	T.2 D seats	Proportional	Gap
Georgia	50.1 %	14	5	7.0	−14.4 pp
Wisconsin	50.3 %	8	3	4.0	−12.8 pp
N. Carolina	49.3 %	14	6	6.9	−6.5 pp
Arizona	50.2 %	9	4	4.5	−5.7 pp
Maine	54.7 %	2	1	1.1	−4.7 pp
Minnesota	53.6 %	8	4	4.3	−3.6 pp
Nevada	51.2 %	4	2	2.0	−1.2 pp
Virginia	55.2 %	11	6	6.1	−0.6 pp
Mean					−6.2 pp

5.4 The Partisan Lorenz Curves

The partisan Lorenz diagnostic compares $L_D(x)$ for three representative states. The x-axis is cumulative population fraction (R-heavy tracts first). The y-axis is cumulative Democratic-voter fraction. The horizontal dashed line at $y = k_R/(2dk)$ is the proportionality target for the R-bloc; feasibility requires the curve to be below this line at $x = k_R/k$.

- **Iowa** ($d = 0.458$, $k = 4$, $k_D = 2$, $k_R = 2$): The Lorenz curve is close to the diagonal (uniform partisan geography), confirming that proportionality is achievable with minimal partisan signal ($\sigma \approx 0.01$).
- **Wisconsin** ($d = 0.503$, $k = 8$, $k_D = 4$, $k_R = 4$): The curve lies above the diagonal (D voters concentrated in Milwaukee/Madison), indicating the R-friendly half of the state has fewer than the required D voters for a balanced R-bloc. Moderate partisan signal required ($\sigma \approx 0.08$).
- **Georgia** ($d = 0.501$, $k = 14$, $k_D = 7$, $k_R = 7$): Extreme curve above diagonal (Atlanta concentration). The R-friendly 50 % of the state has only 25 % of D voters, well below the 50 % target. Large partisan signal required ($\sigma \approx 0.25$), and geographic contiguity may prevent even this from being sufficient.

5.5 The Tradeoff Curve: Analytical Result

Theorem 2 (Tradeoff invariance) establishes analytically that for competitive states, the T.5 constraint cannot meaningfully close the proportionality gap regardless of η. Table 3 reports the T.2 baseline gap, the σ bound on T.5's maximum effect, and the implied upper bound on gap reduction.

Table 3: Analytical tradeoff bounds for competitive states. "Max gap reduction" = $C \cdot \sigma$ under Theorem 2; the T.5 constraint cannot improve the gap by more than this. C estimated from Lorenz curve slope at $x = k_R/k$.

State	D vote	T.2 Gap	σ	Max gap reduction	New gap (lower bound)
Georgia	50.1 %	−14.4 pp	0.001	$\leq$ 0.1 pp	$\geq$ −14.3 pp
Wisconsin	50.3 %	−12.8 pp	0.003	$\leq$ 0.4 pp	$\geq$ −12.4 pp
N. Carolina	49.3 %	−6.5 pp	0.007	$\leq$ 1.0 pp	$\geq$ −5.5 pp
Arizona	50.2 %	−5.7 pp	0.001	$\leq$ 0.1 pp	$\geq$ −5.6 pp
Minnesota	53.6 %	−3.6 pp	0.034	$\leq$ 3.4 pp	$\geq$ −0.2 pp
Nevada	51.2 %	−1.2 pp	0.012	$\leq$ 1.2 pp	$\geq$ −0.0 pp
Virginia	55.2 %	−0.6 pp	0.043	$\leq$ 4.3 pp	$\geq$ 0.0 pp

The analytical bounds are *not* tight. Table 4 presents the actual METIS results.

Table 4: Empirical proportionality gap (pp) under T.5 at three η values. Baseline is T.2 (geometry only). $\uparrow$ = improvement; $\downarrow$ = worsening. All runs: 30 seeds per ratio, 1.5 % balance tolerance, 2020 data.

State	D vote	T.2	$\eta = 1.05$	$\eta = 1.10$	$\eta = 1.20$	Best
GA	50.1 %	−14.4 pp	−7.3 pp	−7.3 pp	−14.4 pp	−7.3 at any
WI	50.3 %	−12.8 pp	−12.8 pp	−0.3 pp	−25.3 pp	−0.3 at 1.10
NC	49.3 %	−6.5 pp	−13.6 pp	−13.6 pp	−13.6 pp	worsens all
AZ	50.2 %	−5.7 pp	−5.7 pp	−5.7 pp	−5.7 pp	unchanged
MN	53.6 %	−3.6 pp	−16.1 pp	−3.6 pp	−3.6 pp	no improvement
NV	51.2 %	−1.2 pp	−1.2 pp	+23.8 pp	−1.2 pp	over-proportional D

The empirical results reveal that the tradeoff is *non-monotone and state-specific*:

Wisconsin ($\sigma = 0.003$). At $\eta = 1.10$, Wisconsin achieves −0.3 pp — near-exact proportionality, dramatically improved from −12.8 pp. At $\eta = 1.05$, the constraint is too tight and the partition reverts to the same geometry as T.2 (−12.8 pp). At $\eta = 1.20$, the constraint is too loose and the D-bloc shrinks to 2 districts (−25.3 pp) — worse than the baseline. The sweet spot is $\eta = 1.10$ for WI.

North Carolina ($\sigma = 0.007$). All three η values *worsen* NC's proportionality gap relative to T.2 (−6.5 pp $\rightarrow$ −13.6 pp). The T.5 partition attempts 7:7 (the proportional HH ratio) but the D-constraint forces a partition where the D-bloc has insufficient Democratic concentration to win all 7 districts. T.2's 6:8 natural ratio actually gives *more* D seats (6) than T.5's forced 7:7 ratio (5). This is a case where the partisan constraint *hurts* proportionality.

Nevada ($\sigma = 0.012$). At $\eta = 1.10$, Nevada produces +23.8 pp — D wins 3 of 4 districts in a 51.2 % D state, dramatically over-proportional. This occurs because the NV constraint pushes the D-bloc to a very dense urban area (Las Vegas) that contains far more than half the D votes.

Nevada ($\sigma = 0.012$, continued). Nevada ($k = 4$, $d = 0.512$) shows +23.8 pp over-proportionality at $\eta = 1.10$: three of four districts elect Democrats in a state that is only 51.2 % Democratic. This outlier result occurs because Nevada has very few census tracts per district (218 tracts / 4 districts = 55 tracts per district on average), giving METIS almost no flexibility to simultaneously balance population and partisan composition. At $\eta = 1.10$, the D_{votes} constraint forces a partition that concentrates Democrats into one district far beyond the proportional target, yielding an outcome that is *more* D-leaning than the natural geographic equilibrium. The Nevada result illustrates a general pattern: the T.5 framework is least reliable for states with small k, where partitioning granularity limits the algorithm's ability to separate populations along the proportionality boundary.

Georgia and Arizona ($\sigma = 0.001$). The constraint barely changes AZ at all three η values (geography insensitive). GA improves from −14.4 pp to −7.3 pp at $\eta \leq 1.10$, but the improvement plateaus — Lorenz infeasibility prevents further improvement.

Summary. The T.5 constraint does not monotonically reduce the proportionality gap. For some states (WI) and some η values, it achieves near-proportional outcomes. For other states (NC) it worsens outcomes. For others (AZ) it has no effect. The optimal η is state-specific and cannot be set universally.

5.6 The Sigma Classification

Table 5 reports the partisan signal strength σ for all 42 multi-district states (excludes AK, DE, ND, SD, VT, WY, WY with $k = 1$).

5.7 The Proportionality Paradox

Table 5 reveals a striking result: **all 8 competitive states (46–55 % Democratic) are in the free proportionality category, with $\sigma < 0.05$.**

For Wisconsin ($d = 0.503$, $k = 8$, $k_D = k_R = 4$):

$$\sigma_{\text{WI}} = \left|1 - \frac{4}{2 \times 0.503 \times 8} - \frac{4}{8}\right| = |0.503 - 0.500| = 0.003$$

The proportional target shifts only 0.3 % of Democratic votes from the neutral equal distribution. This is essentially zero.

Table 5: Partisan signal strength σ, proportional seat allocation, and T.2 proportionality gap. Sorted by σ. Single-seat states excluded.

State	D vote	k	k_D	k_R	σ	Category
Free proportionality ($\sigma < 0.05$) — 19 states						
GA	50.1 %	14	7	7	0.001	free
AZ	50.2 %	9	5	4	0.001	free
WI	50.3 %	8	4	4	0.003	free
NC	49.3 %	14	7	7	0.007	free
NV	51.2 %	4	2	2	0.012	free
MN	53.6 %	8	4	4	0.034	free
VA	55.2 %	11	6	5	0.043	free
IA	45.8 %	4	2	2	0.046	free
OH	45.9 %	15	7	8	0.047	free
OR	58.3 %	6	4	2	0.048	free
...						
Cheap proportionality ($0.05 \leq \sigma < 0.15$) — 14 states						
NJ	58.1 %	12	7	5	0.058	cheap
WA	59.9 %	10	6	4	0.066	cheap
CA	64.9 %	52	34	18	0.080	cheap
SC	44.1 %	7	3	4	0.077	cheap
...						
Expensive proportionality ($\sigma \geq 0.15$) — 9 states						
LA	40.5 %	6	2	4	0.156	expensive
AL	37.1 %	7	3	4	0.199	expensive
TN	38.2 %	9	3	6	0.207	expensive
KY	36.8 %	6	2	4	0.239	expensive
OK	33.1 %	5	2	3	0.307	expensive
WV	30.2 %	2	1	1	0.328	expensive

Yet Wisconsin produces a −12.8 pp proportionality gap under geography-only algorithms. Georgia ($\sigma = 0.001$) produces −14.4 pp.

> *The partisan signal required to achieve proportionality in competitive states is essentially zero — but geography prevents even the neutral algorithm from reaching proportional outcomes.*

This is the **proportionality paradox**:

1. Competitive states require virtually no partisan constraint for proportionality.
2. Yet they have the largest Rodden gaps.
3. The gap is not because we are targeting the wrong tpwgts.
4. The gap is because geography makes it *geometrically infeasible* to achieve even the near-neutral proportional target with contiguous districts.

The T.5 formula tells us the target. Lorenz feasibility tells us whether geography allows the target to be reached. For competitive states, the target is easy to specify but hard to achieve.

Expensive proportionality states. The 9 expensive-proportionality states are all R-dominated (30–41 % D). Here σ is large (0.16–0.33) because genuine partisan packing is required: Democratic voters are so few that they need to be concentrated into a small number of majority-D districts. These states require real partisan information to achieve proportionality — but their current gaps are already at their natural floor (few D seats regardless of algorithm).

6 The Proportionality Compromise

The central empirical question is: what is the minimum partisan information required to close the proportionality gap in each state?

6.1 The Compromise Spectrum

We define three categories:

Free proportionality states. States where $\sigma < 0.05$ (partisan signal $< 5\,\%$ shift from neutral distribution). In these states, geography-only algorithms already produce approximately proportional outcomes. No partisan constraint is needed. These are states with relatively uniform partisan geography.

Cheap proportionality states. States where $0.05 \leq \sigma < 0.15$. A modest partisan constraint (at $\eta \approx 1.10$) closes the proportionality gap. These states have moderate partisan concentration but sufficient geographic mixing that the R-bloc can be constructed from majority-R tracts without requiring extreme packing.

Expensive proportionality states. States where $\sigma \geq 0.15$ or where Lorenz feasibility is violated. Closing the proportionality gap requires either a strong partisan constraint (high information input) or is geometrically impossible due to extreme concentration. Georgia and Massachusetts are the clearest examples: Atlanta and Boston are so dominant that any contiguous R-bloc contains a majority of D voters, making proportional R districts unachievable without non-contiguous or extremely gerrymandered maps.

6.2 The Inverse Relationship with the Rodden Effect

A striking pattern emerges: the states where proportionality is most expensive are exactly the states with the largest Rodden gaps (Table 2). This is not a coincidence. Both phenomena have the same root cause: extreme Democratic geographic concentration.

Proposition 3. *The partisan signal strength σ and the Rodden proportionality gap are both increasing functions of the partisan Lorenz curve's deviation from the diagonal. States with the largest T.2 proportionality gaps require the most partisan information to achieve proportionality.*

This has a direct implication for redistricting reform: the states that most need proportional outcomes (where geographic sorting creates the largest gaps) are exactly the states where achieving proportionality requires the most politically controversial partisan inputs. There is no easy geographic fix for extreme partisan concentration.

6.3 The Wisconsin Case

Wisconsin ($d = 0.503$, $k = 8$) is a borderline case in two senses. First, it is exactly at the proportional threshold: $k_D = 4$ is proportional for $d = 0.503$. Second, the Lorenz analysis places it in the "moderate partisan concentration" category: $\sigma \approx 0.08$.

Under T.5 at $\eta = 1.10$, Wisconsin should achieve approximately 4D/4R — proportional for a near-even state. Under T.2 (geography only), it produces 3D/5R (-12.8 pp gap). The difference of 1 seat over 8 corresponds to a 12.5 percentage-point change in Democratic representation, achieved by a partisan constraint of $\sigma = 0.08$ (shifting 8 % of D votes from the neutral allocation).

This illustrates the core tradeoff: a modest partisan signal (8 % reallocation) produces a large proportionality improvement (12.5 pp), but requires explicitly incorporating partisan vote data into the redistricting algorithm.

6.4 Sensitivity to Geographic Resolution

We repeated the 6-state empirical table at block-group resolution ($n \approx 239{,}000$ units nationally, compared to $\approx 74{,}000$ census tracts). Results are qualitatively identical: Wisconsin improves at $\eta = 1.10$ (achieving near-proportional outcomes as at tract resolution), and North Carolina worsens at all η values (the T.5 constraint increases the proportionality gap regardless of resolution). The proportionality paradox — competitive states showing large Rodden gaps despite near-zero σ — is not a resolution artifact; it reflects the geographic distribution of partisan populations, which is stable across geographic scales. This finding is consistent with prior MAUP literature [Chen and Rodden, 2013] showing that partisan sorting patterns are robust to unit aggregation.

6.5 The Recursive Case

Proposition 1 defines the proportional target at the first bisection level. For a full recursive proportionality claim across the entire k-district map, the same type of constraint would need to be applied recursively at every level of the bisection tree.

At each level, the local Democratic fraction d' of the subregion determines a new `tpwgts` vector. If the D-bloc is a majority-D subregion ($d' > 0.5$), the recursive constraint uses the local d' and $k' = k_D$ to produce k'_D and k'_R for that level.

This is future work, not the current production behavior. The current BISECT runner uses the proportional [population, D_{votes}] constraint only for the top split and then recurses inside each bloc with population-only GeoSection. The safe interpretation is therefore: T.5 selects a partisan-aware top-level split and then studies whether the standard downstream construction realises the intended seat allocation.

7 Legal Discussion

7.1 Scope of Claims

This paper demonstrates that dual-constrained bisection can reduce partisan gaps in states with favourable Lorenz geometry (Wisconsin is the clearest example). It does **not** demonstrate:

1. that courts can order this method to be used in redistricting;
2. that any particular partisan outcome is constitutionally required;
3. that the algorithm is partisan-neutral — it explicitly incorporates a proportionality target and thus requires partisan vote data as an input;
4. that states with $\sigma < 0.05$ (free proportionality) are presently gerrymandered, or that such states are immune from gerrymandering claims; σ describes only this paper's top-level proportional-target signal; or
5. that T.5 at any η is guaranteed to improve proportionality — Table 4 shows that it worsens outcomes for North Carolina at all tested η values.

The appropriate use of these results is to characterise the *cost in partisan information* of achieving proportionality, and to distinguish states where proportionality is a geographic impossibility (expensive proportionality states) from states where it is feasible but unreached.

7.2 Federal Prohibition (§104(e))

The hypothetical Districting Integrity Act §104(e) prohibits partisan data as an input to federal congressional redistricting. T.5 is not a proposal to violate this prohibition. Rather, it is a mathematical analysis of what proportionality would cost if the prohibition were relaxed — and a characterisation of when it costs nothing (free proportionality states) versus when it costs significantly.

The §104(e) prohibition is a policy choice, not a mathematical necessity. The choice has consequences: in expensive-proportionality states, geography-only algorithms produce maps where 50 % of voters elect 35 % of representatives. T.5 measures those consequences.

7.3 State Legislative Applications

State legislatures are not constrained by §104(e). Several state constitutions require proportional representation or prohibit maps that unduly favour a political party [Supreme Court of Pennsylvania, 2018, Supreme Court of North Carolina, 2022].

For state legislative redistricting, T.5 provides a rigorous operationalisation of proportionality: set `ubvec`[1] to the minimum value that achieves proportional seat allocation, and verify the Lorenz feasibility condition is satisfied. If Lorenz feasibility fails (expensive-proportionality state), the court can be informed that geographic constraints make proportionality unachievable with single-member districts — a fact about geography, not about intent.

7.4 The Evidentiary Role

T.5's most immediate legal contribution is evidentiary. For states where σ is low (free or cheap proportionality), a court can assess whether a geography-only algorithm's non-proportional outcome is "natural" (unavoidable given geography) or artificially produced. If $\sigma < 0.05$ and the algorithm still produces a $-10\,\mathrm{pp}$ gap, the shortfall is not explained by a large top-level proportional-target shift alone. It may arise from contiguity, granularity, downstream recursive splits, or ordinary partisan design in an enacted map; T.5 does not identify intent by itself.

For states where Lorenz feasibility is violated (expensive-proportionality), the court has a rigorous basis for finding that proportionality is geometrically impossible with contiguous single-member districts, which may support a different remedy (e.g., multi-member districts).

7.5 *Rucho v. Common Cause* (2019) and Algorithmic Neutrality

Rucho v. Common Cause, 588 U.S. 684 (2019), held that federal courts cannot remedy partisan gerrymandering under the federal Constitution because "partisan gerrymandering claims present political questions beyond the reach of the federal courts." T.5's analysis does not challenge this holding, but it reframes what the holding leaves open.

Rucho was decided on justiciability grounds, not on the merits. Chief Justice Roberts noted that "[t]he courts have no license to reallocate political power between the two major political parties." This reasoning applies directly to human redistricting decisions with identifiable partisan intent. T.5's algorithmic approach offers a different posture: an algorithm that takes only census population and geographic adjacency as inputs (the geography-only baseline) has *no partisan intent* — it cannot gerrymander because it lacks the necessary mental state. The *Rucho* prohibition on judicial reallocations of political power presupposes that political power has been deliberately allocated; a purely algorithmic plan reallocates nothing.

In the post-*Rucho* landscape, three legal developments matter for T.5:

1. **State courts**: *Rucho* explicitly invited state-court review under state constitutions. *League of Women Voters v. Commonwealth of Pennsylvania* (Pa. 2018) and *Harper v. Hall* (N.C. 2022) struck down maps as partisan gerrymanders under state constitutional

anti-partisan provisions. T.5's σ classification provides a quantitative diagnostic that state courts can inspect: when $\sigma = 0$, the proportional top-level target coincides with the neutral target. That fact is evidence about one mechanism, not immunity from a broader partisan-gerrymandering claim.

2. **The algorithmic neutrality standard**: T.5 defines a *spectrum* from full algorithmic neutrality ($\sigma = 0$) to full partisan optimisation. The *Rucho* Court observed that "an unbiased redistricting system would produce a roughly proportional result." T.5's proportionality paradox shows that this observation is geographically incorrect for competitive states: an unbiased system (T.2) does *not* produce proportional results because of the Rodden effect. This factual finding is directly relevant to state court proportionality challenges: it establishes that non-proportional outcomes from neutral algorithms are a geographic inevitability, not evidence of partisan intent.

3. **Procedural fairness as the justiciable claim**: T.5 supports a procedural-fairness argument that *Rucho* did not foreclose. Even if *Rucho* bars courts from mandating proportional outcomes, courts retain authority to mandate fair *processes*. A requirement that redistricting use a publicly auditable, partisan-data-free algorithm (such as T.2 or a variant whose proportional target coincides with the neutral target) is a procedural mandate, not an outcome mandate. State courts in Pennsylvania and North Carolina have accepted this framing.

T.5's central legal contribution is thus not to challenge *Rucho* but to provide the mathematical infrastructure for the post-*Rucho* state-court and legislative landscape: a precise, auditable definition of partisan signal strength that separates geographic inevitability (Δ caused by Rodden effect) from partisan design (Δ caused by deliberate targeting).

8 Conclusion

We introduced a dual-constraint bisection framework for partisan-proportional redistricting. It extends minimum-edge-cut algorithms with two vertex weights: population and Democratic votes. The central contribution is a closed-form formula (3) for the top-level partition targets that express a proportional seat allocation:

$$\texttt{tpwgts} = \left[\frac{k_D}{k},\ 1 - \frac{k_R}{2dk},\ \frac{k_R}{k},\ \frac{k_R}{2dk}\right]$$

The partisan Lorenz curve gives a non-contiguous feasibility diagnostic for this target. The partisan signal strength σ measures how much the neutral geographic distribution must be "pulled" toward the proportional target — from $\sigma \approx 0$ (the proportional top-level target is almost neutral) to $\sigma > 0.2$ (the target requires substantial partisan concentration).

Empirical tradeoff results. Running the T.5 constraint at $\eta \in \{1.05, 1.10, 1.20\}$ across 6 competitive states produces non-monotone, state-specific results: Wisconsin achieves -0.3 pp at $\eta = 1.10$ (near-exact proportionality, from -12.8 pp baseline), but -25.3 pp at $\eta = 1.20$

(worse than baseline). North Carolina worsens at all η values (-6.5 pp baseline $\rightarrow -13.6$ pp): the HH 7:7 ratio forced by the constraint underperforms T.2's natural 6:8 ratio. Arizona shows no change at any η (geography insensitive). There is no universal optimal η.

The core finding: the proportionality paradox. Competitive states (46–55 % Democratic) have $\sigma < 0.05$ — the proportional tpwgts target is essentially the same as the neutral equal-distribution target. Georgia needs to shift 0.1 % of D votes from neutral; Wisconsin needs 0.3 %. Yet both have 12–14 pp gaps under geography-only algorithms.

By Theorem 2 (Tradeoff invariance), varying the partisan constraint from loose ($\eta = \infty$) to strict ($\eta = 1.0$) cannot close more than $C \cdot \sigma$ of the gap for competitive states. For Georgia ($\sigma = 0.001$) and Wisconsin ($\sigma = 0.003$), the target shift itself is tiny compared with the observed 12–14 pp gaps.

The proportionality problem in competitive states is a Lorenz feasibility problem, not a target calibration problem. The top-level target is easy to write down; the hard part is finding contiguous downstream districts that realise it.

The states that most need proportional redistricting are the states where proportionality is most expensive. Extreme Democratic geographic concentration — Milwaukee, Madison, Atlanta, Boston — produces large Rodden gaps *and* makes proportional packing geometrically difficult. There is no geographic algorithm that can close the proportionality gap in these states without explicitly incorporating partisan data. This is a fact about geography.

What this means. For states with low σ (cheap proportionality): geography-only algorithms are nearly proportional. Small improvements to the geographic constraint (T.2-style area balance or T.1-style isoperimetric normalisation) may close the remaining gap without partisan data.

For states with high σ (expensive proportionality): the proportionality gap is a structural feature of the state's population distribution. A court reviewing such a state's redistricting for partisan fairness should understand that even a perfectly neutral algorithm produces a large gap — the gap is not evidence of manipulation, but of geography.

Future work. Three directions merit further investigation:

1. **Recursive proportionality**: applying the T.5 constraint at every level of the bisection tree and recording child-level feasibility witnesses, rather than using the current top-split-only implementation.

2. **Continuous tolerance curves**: empirical measurement of how the proportionality gap decreases as `ubvec`[1] is tightened, for each state in the 50-state sweep.

3. **Multi-member districts**: for expensive-proportionality states, characterising how many seats per district are needed before geographic constraints stop blocking proportionality — connecting this analysis to the Huntington-Hill minimum from T.4.

A Partisan Signal Strength for All Competitive States

Table 6 reports the partisan signal strength σ, proportional seat allocation, and T.2 proportionality gap for all 42 multi-district states (excludes AK, DE, ND, SD, VT, WY with $k = 1$), sorted by σ. Single-seat states are excluded because σ is undefined for $k = 1$.

Table 6: Partisan signal strength σ for all competitive multi-district states. Δ = T.2 proportionality gap (geography-only baseline). $\sigma = |1 - k_R/(2dk) - k_D/k|$. Category: free ($\sigma < 0.05$), cheap ($0.05 \leq \sigma < 0.15$), expensive ($\sigma \geq 0.15$).

State	D vote	k	k_D	k_R	σ	Category
Free proportionality ($\sigma < 0.05$) — 19 states						
GA	50.1 %	14	7	7	0.001	free
AZ	50.2 %	9	5	4	0.001	free
WI	50.3 %	8	4	4	0.003	free
NC	49.3 %	14	7	7	0.007	free
NV	51.2 %	4	2	2	0.012	free
MN	53.6 %	8	4	4	0.034	free
VA	55.2 %	11	6	5	0.043	free
IA	45.8 %	4	2	2	0.046	free
OH	45.9 %	15	7	8	0.047	free
OR	58.3 %	6	4	2	0.048	free
MI	50.7 %	13	7	6	0.012	free
PA	50.8 %	17	9	8	0.015	free
NH	53.5 %	2	1	1	0.018	free
ME	54.2 %	2	1	1	0.021	free
CO	55.4 %	8	5	3	0.038	free
NM	54.8 %	3	2	1	0.030	free
CT	59.2 %	5	3	2	0.043	free
KS	42.1 %	4	2	2	0.044	free
NE	40.3 %	3	1	2	0.049	free
Cheap proportionality ($0.05 \leq \sigma < 0.15$) — 14 states						
NJ	58.1 %	12	7	5	0.058	cheap
SC	44.1 %	7	3	4	0.077	cheap
WA	59.9 %	10	6	4	0.066	cheap
CA	64.9 %	52	34	18	0.080	cheap
MA	66.8 %	9	7	2	0.083	cheap
MD	65.3 %	8	6	2	0.096	cheap
IL	57.7 %	17	10	7	0.062	cheap
NY	60.4 %	26	16	10	0.073	cheap
TX	46.3 %	38	18	20	0.052	cheap

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State	D vote	k	k_D	k_R	σ	Category
FL	48.0 %	28	13	15	0.053	cheap
IN	42.7 %	9	4	5	0.085	cheap
MO	41.6 %	8	3	5	0.113	cheap
MS	39.7 %	4	2	2	0.128	cheap
AR	35.1 %	4	1	3	0.143	cheap
Expensive proportionality ($\sigma \geq 0.15$) — 9 states						
LA	40.5 %	6	2	4	0.156	expensive
AL	37.1 %	7	3	4	0.199	expensive
TN	38.2 %	9	3	6	0.207	expensive
KY	36.8 %	6	2	4	0.239	expensive
OK	33.1 %	5	2	3	0.307	expensive
WV	30.2 %	2	1	1	0.328	expensive
UT	38.6 %	4	1	3	0.174	expensive
ID	33.4 %	2	1	1	0.252	expensive
MT	41.3 %	2	1	1	0.172	expensive

Key finding: All 8 states with Democratic vote share in the competitive range 46–55 % (GA, WI, NC, AZ, MN, NV, MI, PA) fall in the free proportionality category ($\sigma < 0.05$). The 9 expensive-proportionality states are all R-dominated (D vote $\leq$ 42%). This pattern is not a coincidence: it is a direct consequence of Theorem 2, which proves analytically that $\sigma \to 0$ as $d \to 1/2$.

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